DRUGS · THYROID AND PARATHYROID

Potassium iodide and preoperative preparation for thyroid surgery

Operating on a thyrotoxic patient risks thyroid storm, a life-threatening surge of thyroid hormone with fever, tachycardia, arrhythmia, heart failure and agitation. Patients are prepared with antithyroid drugs and beta blockers, and potassium iodide (as Lugol's iodine or tablets) is given for about 10 days before thyroidectomy for Graves' disease to reduce thyroid hormone release and gland vascularity. Thyroid storm itself is treated with a sequence of drugs given in a particular order.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

PREOPERATIVE PREPARATION FOR THYROID SURGERY

	SUMMARY
Potassium iodide	Lugol's iodine (aqueous iodine oral solution, about 130 mg/ml total iodine) 0.1 to 0.3 ml three times daily in milk or water, or potassium iodide tablets, for 7 to 10 days before surgery in Graves' disease
How iodine works	High iodide concentrations acutely inhibit hormone synthesis (Wolff-Chaikoff effect) and release, and reduce gland vascularity; the effect wanes after 10 to 14 days (escape)
Beta blockers	Propranolol 40 mg three or four times daily (up to 80 mg four times daily) to control heart rate and tremor; also reduces peripheral T4 to T3 conversion at high doses; use cardioselective agents or diltiazem in asthma
Steroids	Hydrocortisone or dexamethasone reduce T4 to T3 conversion and treat relative adrenal insufficiency in storm
Cholestyramine	Binds thyroid hormone in the gut and reduces enterohepatic recirculation; an adjunct for rapid preparation

	SUMMARY
Plasma exchange	For refractory cases before urgent surgery

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Elective surgery

The patient should be clinically and biochemically euthyroid on carbimazole (or PTU), with beta blockade for symptoms. Iodine is then added for the last 7 to 10 days in Graves' disease. Toxic nodular goitre does not usually need iodine.

2.2 Thyroid storm

Recognise it by fever, tachycardia or atrial fibrillation, heart failure, agitation or confusion and gastrointestinal upset (the Burch-Wartofsky score helps). Treat in critical care:

THYROID STORM

1. **Block synthesis**
   - → PTU (loading 500 to 1,000 mg, then 250 mg every 4 hours) or carbimazole or methimazole
2. **Block release**
   - → iodine, given at least an hour after the thionamide, so it is not used to make more hormone
3. **Block peripheral effects**
   - → propranolol (or esmolol)
   - → hydrocortisone 300 mg then 100 mg every 8 hours
4. **Support**

→ cooling, fluids, treatment of heart failure and arrhythmias, paracetamol (not aspirin, which displaces T4 from binding proteins)

→ find and treat the trigger

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Iodine: metallic taste, sialadenitis and salivary swelling ("iodide mumps"), rash, gastrointestinal upset, hypersensitivity. Beta blockers: bronchospasm, bradycardia, heart failure.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The **Wolff-Chaikoff** effect and escape.
- Preoperative preparation in Graves' disease.
- The order of treatment in **thyroid storm**.

4.2 Common pitfalls

COMMON PITFALLS

- Giving iodine before the thionamide in thyroid storm.

- Proceeding with elective surgery in a thyrotoxic patient.

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